

Demystifying the Cloud

A Beginner's Guide to SaaS, PaaS, and IaaS



SaaS

Software as a Service



PaaS

Platform as a Service



IaaS

Infrastructure as a Service

The Great Transformation

IT is shifting from a product you own to a service you consume



The Electricity Analogy

In the early industrial revolution, every business built its own power generator — they owned the product. Eventually, the electric grid appeared and electricity became a service: plug in and pay for what you use. Cloud computing applies this exact logic to digital resources.

BEFORE — IT as a Product

- Buy physical servers upfront
- Purchase & maintain software licenses
- Pay for idle capacity 24/7
- Slow to scale — weeks of procurement

AFTER — IT as a Service

- Rent computing power on-demand
- Access software via browser
- Pay only for what you use
- Scale globally in minutes

SaaS

Software
as a Service



The User's Layer

What is SaaS?

Software delivered over the internet — no local installation required. The vendor hosts, maintains, and updates everything. You access it through a browser or mobile app.

Real-World Examples

- Webmail (Gmail, Outlook) — access inbox from anywhere
- Productivity tools (Google Docs, Microsoft 365)
- CRM & ERP systems (Salesforce, SAP)

Top 3 Benefits



Global Availability

Access from any device with a network connection



No Manual Updates

Vendor handles all maintenance and patches



Browser-Based

No complex installation — browser is the interface

PaaS

Platform
as a Service



The Developer's Toolset

What is PaaS?

A pre-configured development environment delivered over the cloud. Developers focus entirely on writing code and logic — the platform handles the OS, runtime, and underlying infrastructure.

"PaaS allows for the rapid response to changing user needs."

Management Split

YOU manage

Your application code & data

VENDOR manages

Runtime, OS, servers, storage, networking

Examples: Heroku · Google App Engine · AWS Elastic Beanstalk

IaaS

Infrastructure
as a Service



The Digital Foundation

What is IaaS?

The most foundational layer. Organisations rent raw data-centre hardware — compute, storage, and networking — rather than buying physical equipment. The Hypervisor partitions hardware into multiple virtual machines (instances).

Primary Shared Resources



CPU

Processing power for
computations



Memory

RAM for active task data



I/O

Data movement in/out of
compute

Responsibility Split

You manage: Guest OS + Applications running inside your instance

Vendor manages: Physical hardware, hypervisor, data centre

The Cloud Service Spectrum

How responsibility shifts across the three models

Model	Primary Example	You Manage	Vendor Manages	Best For
SaaS	Gmail, Salesforce, Microsoft 365	Nothing — Vendor handles all	Entire application, OS, hardware	End users needing ready-made apps
PaaS	Heroku, Google App Engine	Application code & data	Platform, OS, runtime, servers	Developers building & deploying apps
IaaS	AWS EC2, Azure VMs, Google GCE	Guest OS & applications	Physical hardware & hypervisor	IT teams needing full control

Key insight: the higher the layer, the more the vendor manages — and the less you control.

The Cloud Advantage

Why elasticity and scalability change everything



Scalability

The capacity to handle increasing workloads by adding resources — more users, more data, more compute.



Elasticity

The ability to scale quickly AND shrink back down. Resources contract when not needed — you never pay for idle hardware.

Simplifying IT Management



Reduced Procurement

Resources available immediately — no weeks of ordering



Pay-As-You-Go

No upfront fees — costs strictly tied to consumption



Greener Computing

Shared data centres reduce overall carbon footprint

Navigating the Challenges

These are learning checkpoints — every architect must plan for them



Migration

Moving existing data and legacy apps into the cloud is technically complex. Requires careful planning to avoid downtime or data loss.



Security & Privacy

Protecting data in a shared, multi-tenant environment. Who can access your data? Are encryption standards adequate?



Vendor Lock-In

Becoming dependent on proprietary tools makes switching providers painful and expensive. Design for portability.



Legal & Compliance

Data sovereignty laws vary by country. Service Level Agreements (SLAs) define your rights when things go wrong.



Internet Dependence

Cloud services rely entirely on network connectivity. Poor or lost connection means lost access to critical systems.

Your Journey Into the Cloud

Key takeaways & where to go next

- Cloud transforms IT from a product you own to a metered service you consume
- SaaS = finished app for users · PaaS = platform for developers · IaaS = raw infrastructure
- Elasticity solves the idle-capacity problem — you only pay for what you actually use
- Every cloud adoption involves real challenges: migration, security, lock-in, and compliance

Get Hands-On

Amazon Web Services

AWS Free Tier — 12 months free

Microsoft Azure

Free \$200 credit to start

Google Cloud Platform

90-day \$300 credit available